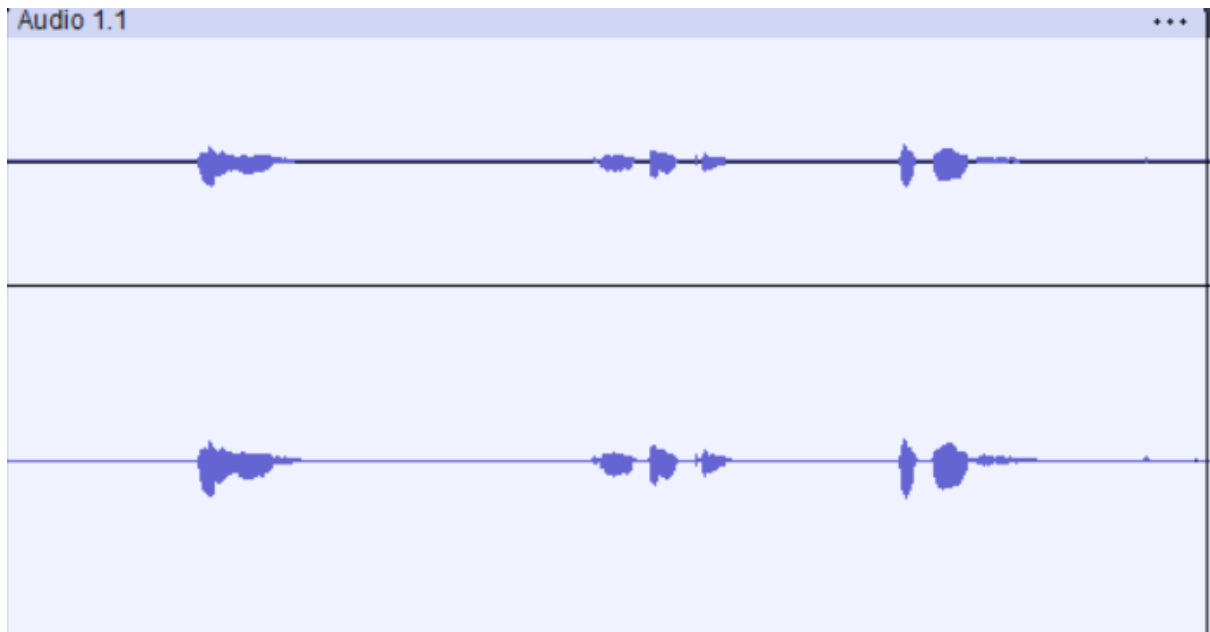


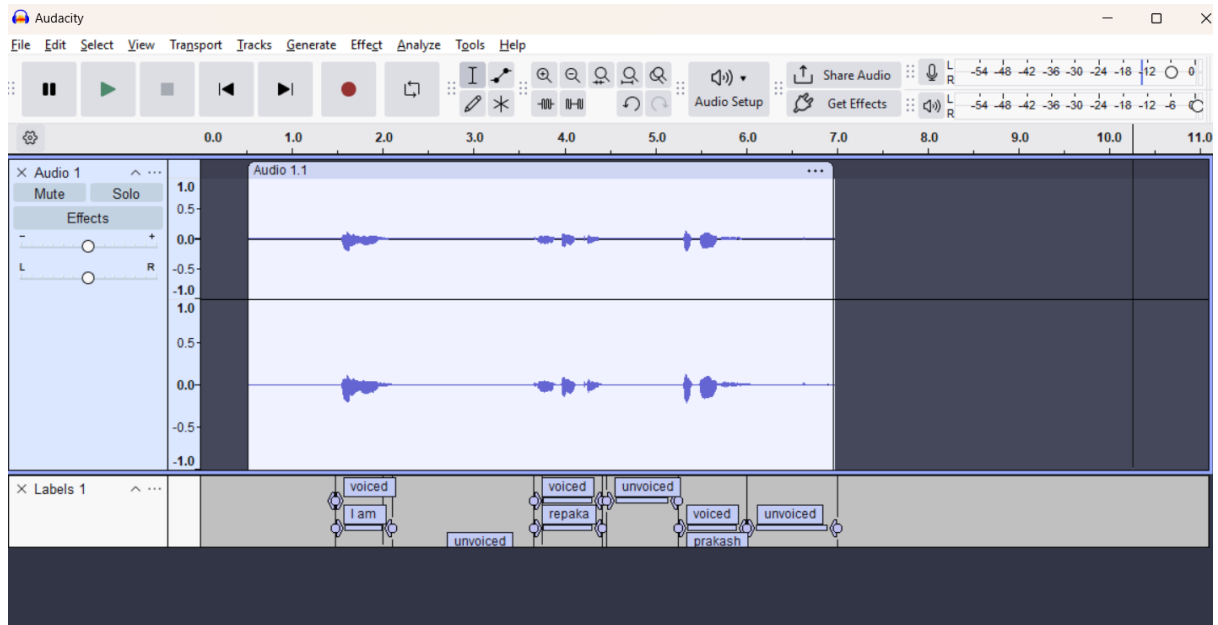
ASSIGNMENT 2

TASK 1

WAVE FORM SENTENCE: I AM REPAKA PRAKASH.

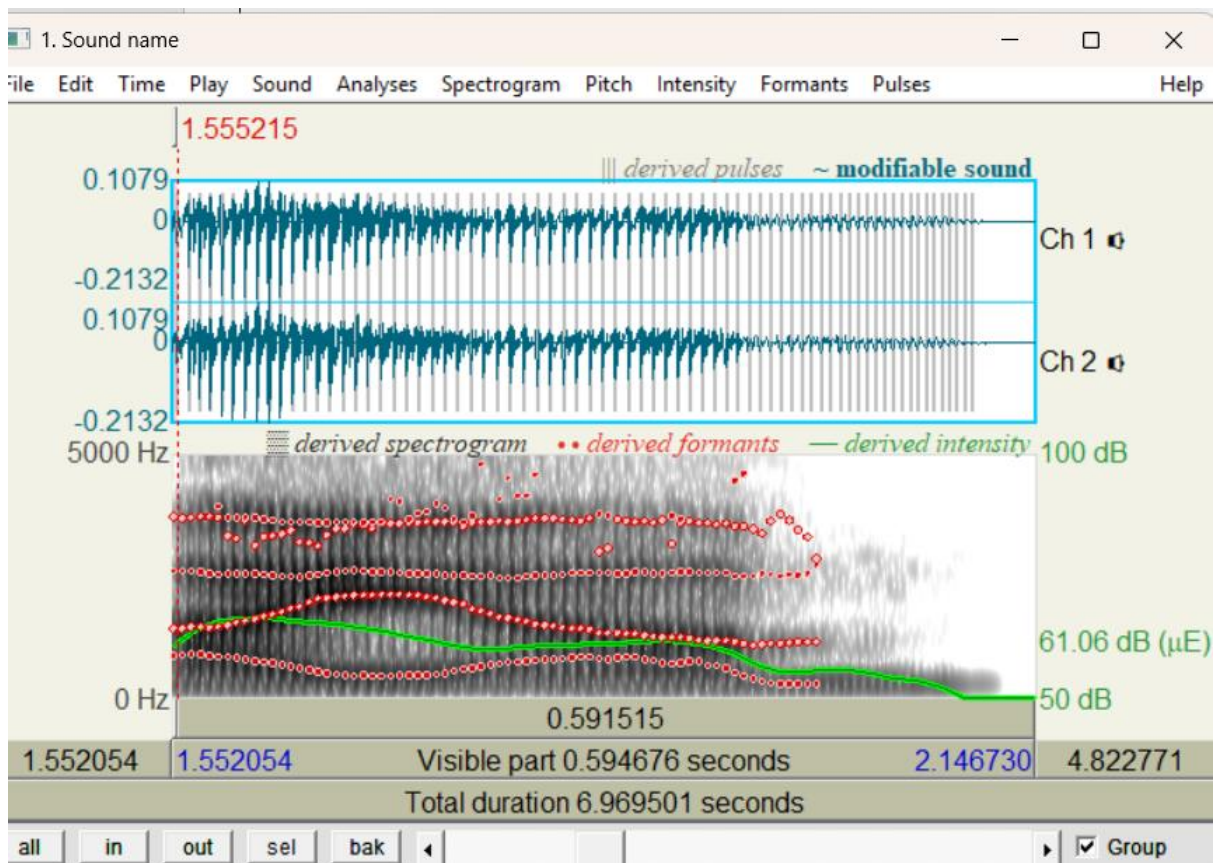
TIME DOMAIN PLOT ON AUDACITY





VOICED & UNVOICED MARKING ON AUDACITY

WAVEFORM OF SELECTED REGION – “IAM”:

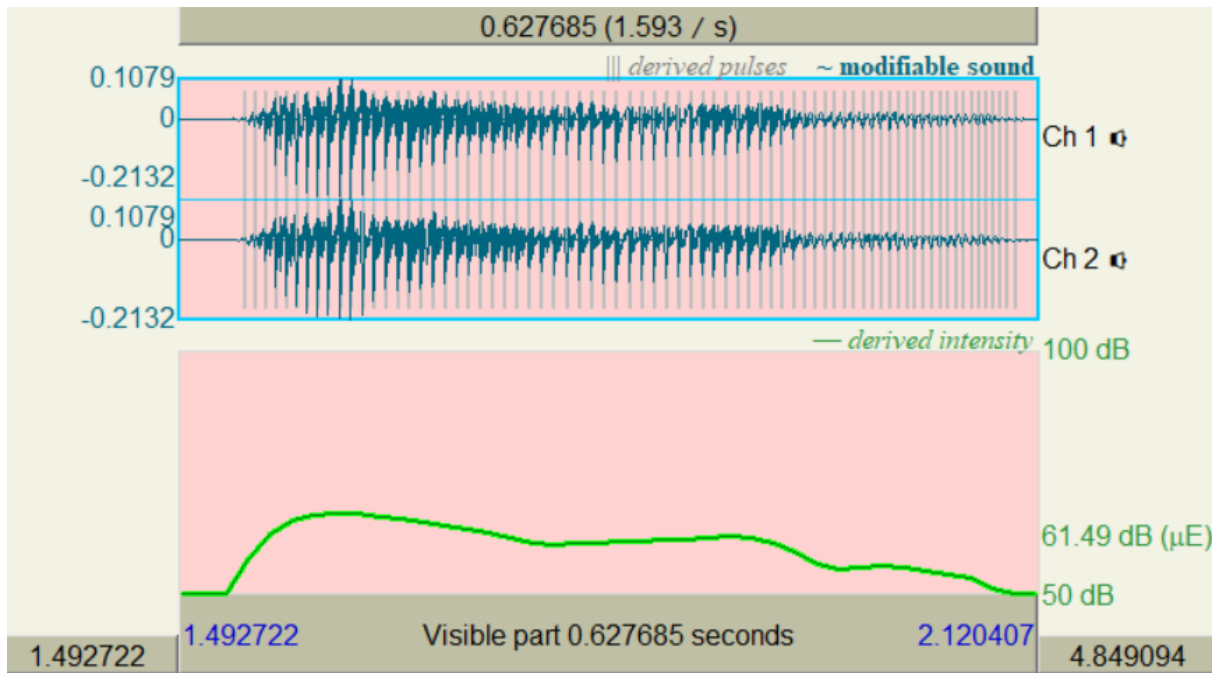


Energy

What it is: The "loudness" or amplitude of the signal.

Observation: The waveform is "tall," reaching high and low amplitude values.

ENERGY / INTENSITY of selected region : 61.658716598619904 dB (mean-energy intensity in SELECTION)mean of selected region)



Voiced segments have **High Energy** because they are produced with significant power from the lungs and vocal folds.

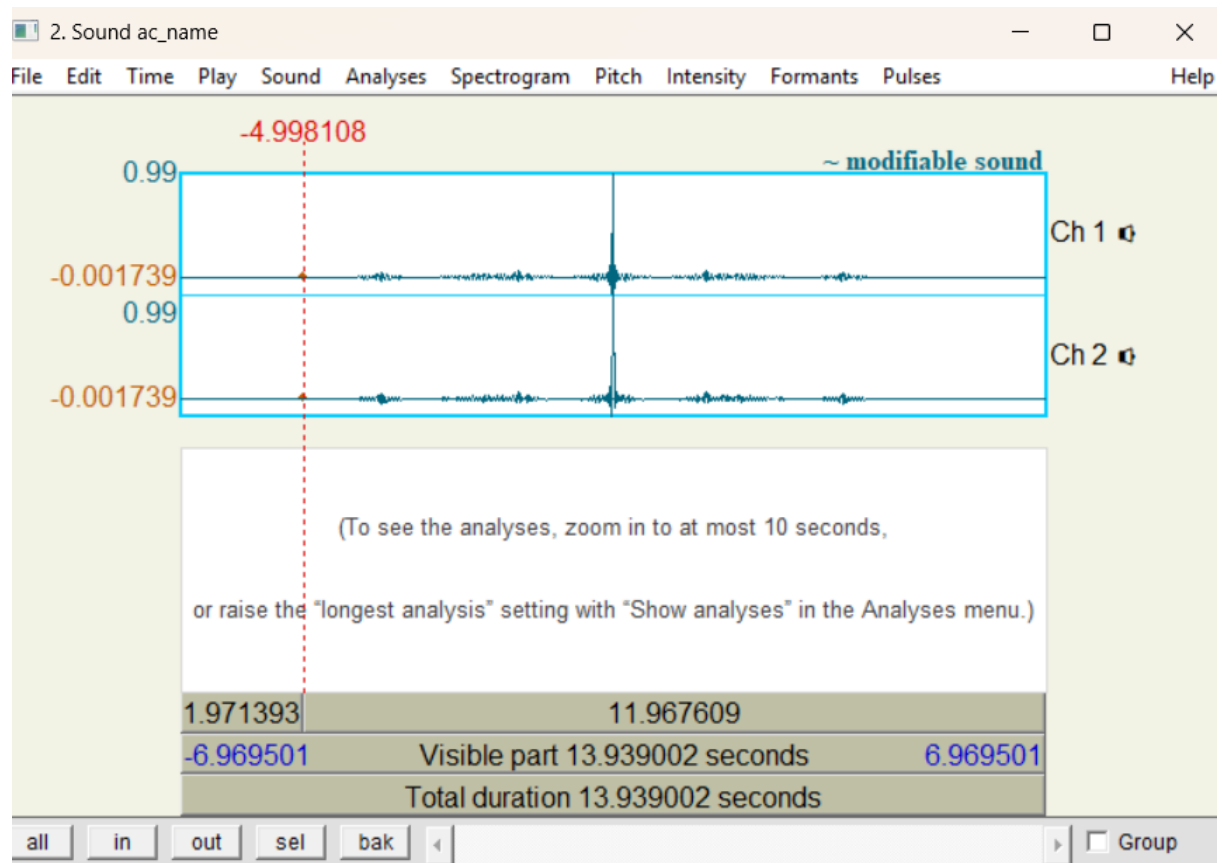
COMMENT: INTENSITY IS RELATIVELY HIGH (79 DB) .This confirms the **high energy** nature of voiced sounds.

Autocorrelation

What it is: A measure of how well the signal matches a delayed version of itself. It's the best way to find periodicity.

Observation: The plot will show a series of clear, repeating peaks.
Autocorrelation is what Praat uses to find pitch.

PITCH: 120Hz

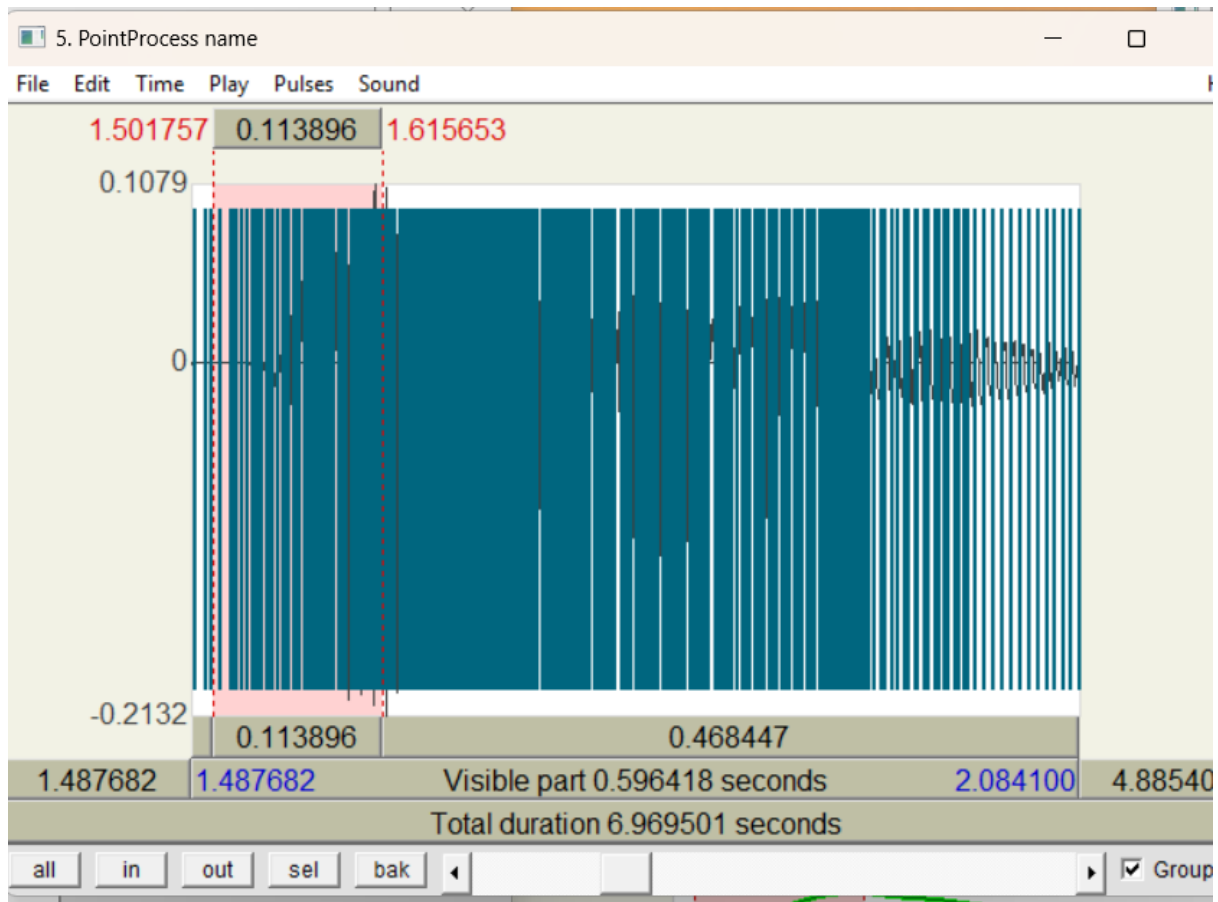


COMMENT: ABOVE 2ND FIGURE -You will see a plot with a strong, clear series of repeating peaks. This is the classic signature of a periodic, voiced sound.

ZERO-CROSSING RATE: ZCR

What it is: How many times the waveform crosses the '0' horizontal line.

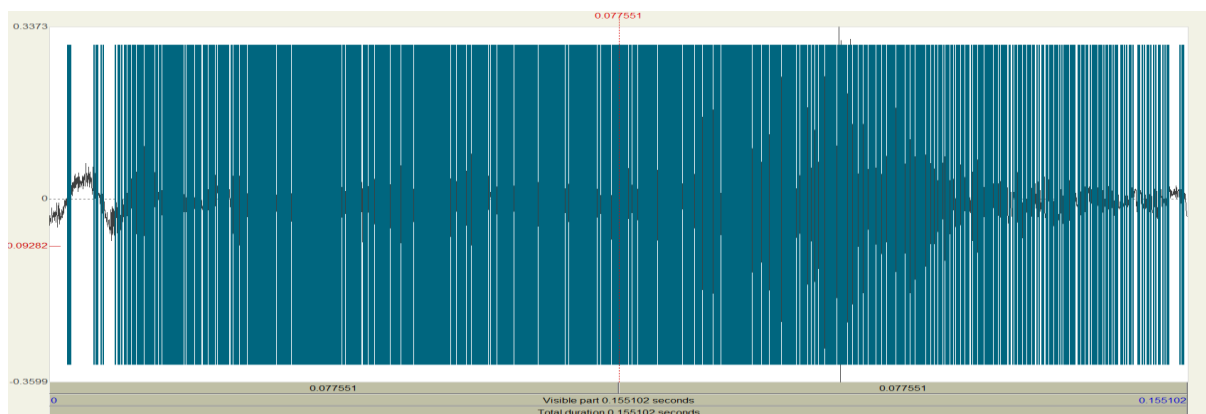
Observation: You'll see the waveform looks like random static, crossing the zero line constantly and erratically.



COMMENT: Voiced segments have a Low ZCR because their energy is concentrated in lower frequencies (the fundamental frequency and its harmonics).

What it is: How many times the waveform crosses the '0' horizontal line.

Observation: Zoom in. You'll see the waveform looks like random static, crossing the zero line constantly and erratically.



Definition: What Are Formants?

In simple terms, formants are the resonant frequencies of your vocal tract.

Think of your voice as a two-part system:

1. The Source: Your vocal folds (vocal cords) vibrate, creating a "buzz" sound. This buzz has a base pitch (F_0) and many harmonics.
2. The Filter: Your throat, mouth, and nasal cavity act as a "filter." By changing the shape of your mouth—using your tongue, jaw, and lips—you change the shape of this filter.

This filter amplifies (makes louder) certain frequencies from the "buzz" and dampens (makes quieter) others. The peaks of amplified frequency that come out in the final sound are the formants.

We label them F_1 , F_2 , F_3 , and F_4 , from lowest frequency to highest.

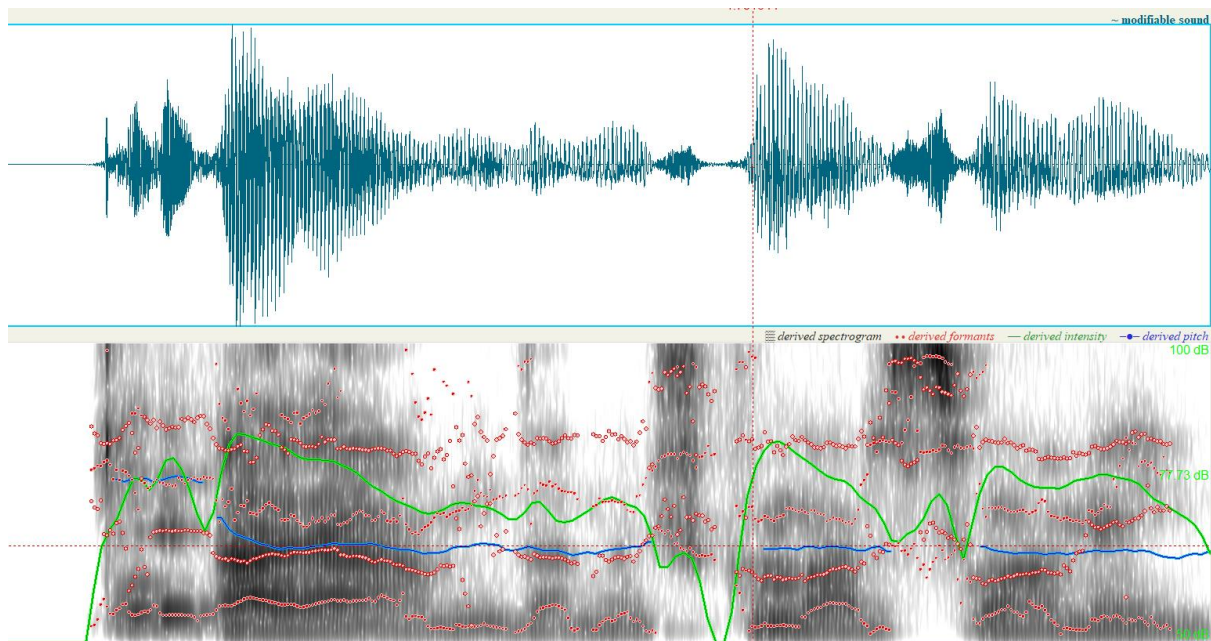
Spectral Details (F_1 , F_2 , F_3)

The first two formants are the most important for identifying *which vowel* you are saying.

- F_1 (First Formant): This is related to vowel height (how open or closed your jaw is).
 - A low F_1 value means a closed or high vowel
 - A high F_1 value means an open or low vowel
- F_2 (Second Formant): This is related to vowel backness (how far forward or back your tongue is).
 - A high F_2 value means a front vowel
 - A low F_2 value means a back vowel
- F_3 (Third Formant): This is often related to "r-coloring" (rhoticity) and can help distinguish sounds like /l/ and /r/. It also adds to the overall timbre or quality of the voice.
- F_4 (Fourth Formant): This formant (and higher ones) are generally considered to be more related to your unique, individual voice quality and "ring" rather than identifying specific vowel sounds.

First, let's identify the main vowel sounds in your sentence: "I am Repaka Prakash". You will be looking for:

1. /aɪ/ (the diphthong in "I")
2. /æ/ (the vowel in "am")
3. /ɑ/ (the vowel in "Repaka," and "Prakash,")
4. /e/ (the vowel in "Repaka")
5. /u/ (the vowel in "Prakash")



/aɪ/ (the diphthong in "I")

Time_s	F1_Hz	F2_Hz	F3_Hz	F4_Hz
1.554099	814.456631	1394.429394	2603.176391	3718.667216
1.560349	873.125516	1421.120099	2596.358593	3677.052957
1.566599	896.312282	1439.337801	2592.306481	3701.288673
1.572849	871.863068	1435.268318	2569.836482	3723.013572
1.579099	836.843143	1430.226275	2530.939746	3701.962366
1.585349	826.236011	1447.827966	2524.953896	3642.019941
1.591599	822.457964	1501.091412	2542.949039	3200.347970
1.597849	810.588388	1555.578753	2546.990569	3446.236852
1.604099	776.257865	1583.335914	2546.254254	3546.893707

1.610349	739.583505	1621.203371	2546.541218	3087.048337
1.616599	724.940045	1676.663963	2529.010737	3337.679820
1.622849	694.660623	1706.327063	2506.820690	3296.026164
1.629099	662.156519	1753.310406	2493.612726	3436.635211
1.635349	620.609431	1827.509477	2488.243526	3329.590367
1.641599	583.955994	1892.335717	2481.253805	3179.118922
1.647849	549.744875	1977.509306	2507.624964	3162.473697
1.654099	521.972703	2025.723475	2540.380777	3161.609280
1.660349	483.133049	2047.837425	2566.893910	3346.355341
1.666599	463.087699	2069.984428	2587.461481	3485.220237
1.672849	448.590231	2081.942020	2602.257186	3499.555403
1.679099	439.459614	2087.462913	2607.453008	3568.501238
1.685349	442.924956	2092.691025	2593.873107	3456.987306
1.691599	442.445626	2097.712448	2585.400586	3489.784017

/æ/ (the vowel in "am")

Time_s	F1_Hz	F2_Hz	F3_Hz	F4_Hz
1.680198	440.626007	2088.195132	2607.474110	3564.139107
1.686448	440.907712	2091.243781	2583.036842	3378.622686
1.692698	441.978733	2098.108590	2578.207086	3422.835705
1.698948	440.756218	2115.827618	2587.185592	3596.416010
1.705198	440.509470	2118.734575	2575.280690	3561.788489
1.711448	446.456875	2106.717966	2548.546025	3574.988946
1.717698	449.877276	2098.148928	2537.299634	3584.240241
1.723948	453.675601	2086.193067	2542.432231	3597.195369
1.730198	465.330561	2047.340361	2543.767448	3605.283838
1.736448	482.494925	2012.897703	2537.982653	3601.993717
1.742698	502.153308	1972.702595	2509.793690	3606.906858
1.748948	553.051202	1939.756223	2503.408884	3507.708048
1.755198	589.229677	1850.764891	2494.302489	3553.488119
1.761448	605.304660	1790.384222	2494.658607	3593.295609
1.767698	649.054766	1744.677022	2502.775970	3610.710658
1.773948	682.217488	1674.231902	2470.945554	3610.095909

1.780198	698.790426	1639.883147	2448.082357	3618.771626
1.786448	720.944599	1604.103136	2448.088387	3616.233047
1.792698	743.505253	1558.858600	2467.527131	3639.849794
1.798948	756.579101	1533.789859	2497.442438	3656.549128
1.805198	767.265724	1497.283934	2508.658541	3683.735785
1.811448	772.482817	1462.322013	2501.088398	3680.801985
1.817698	786.564544	1444.145693	2522.980137	3681.117989
1.823948	798.558614	1418.471187	2538.486336	3650.257498
1.830198	825.287075	1412.505143	2566.569996	3625.142257
1.836448	844.135986	1404.227950	2560.619600	3635.955185
1.842698	796.690620	1364.000380	2556.005277	3723.350306
1.848948	777.753426	1309.557103	2556.165658	2886.277830
1.855198	795.700535	1313.601731	2550.181662	3234.513333
1.861448	809.675134	1311.067159	2548.726123	3633.793287
1.867698	845.837541	1303.683245	2555.064925	3609.676789
1.873948	804.460857	1269.604996	2584.790095	3221.056858
1.880198	769.017527	1257.400434	2573.303182	3627.817443
1.886448	732.737974	1238.918583	2545.794869	3576.635461
1.892698	696.396967	1226.113348	2543.855734	3264.565728
1.898948	694.825212	1220.075738	2564.741451	3317.177337
1.905198	703.834189	1218.113915	2576.591841	3644.342104
1.911448	739.717963	1188.937843	2569.033806	3604.423387
1.917698	704.206468	1168.492682	2551.978628	3590.513653
1.923948	647.389868	1156.014077	2562.707317	3558.083885
1.930198	636.294220	1156.988948	2567.083635	3561.128433
1.936448	567.631215	1143.776133	2531.278692	3558.165119
1.942698	462.356931	1121.282783	2480.761059	3508.465244
1.948948	408.011073	1083.085350	2486.615879	3520.934563
1.955198	350.294532	1067.022475	2514.935227	3394.176716
1.961448	311.807736	1087.782438	2552.627289	3481.468907
1.967698	278.510779	1103.941963	2495.935066	3716.595684
1.973948	275.164868	1096.808328	2508.140366	3744.269500

1.980198	276.946122	1129.457456	2527.819257	3569.688475
1.986448	272.977889	1133.654849	2497.150774	3328.924982
1.992698	278.193722	1147.880537	2601.913896	3234.441412
1.998948	279.906774	1121.287098	2679.707102	2697.767025
2.005198	283.199765	1108.230153	2522.632958	3167.275658
2.011448	281.748966	1077.721514	2537.513365	3527.718625
2.017698	278.800545	919.127803	2486.348480	3347.295149
2.023948	278.201907	993.066599	2296.840977	2946.862934
2.030198	267.106434	1069.093066	2272.059170	2904.196291
2.036448	261.918919	1021.130310	2389.159026	2834.620379
2.042698	273.616067	958.871506	2444.037463	2752.758227
2.048948	276.385171	832.232581	2341.939385	3033.429202
2.055198	282.562754	2341.138680	2520.087884	3739.635572
2.061448	225.344733	1784.138008	2619.009808	3948.230396
2.067698	194.458987	1121.147132	2635.067891	4166.179084
2.073948	188.526887	1491.286429	2611.095030	4329.442305
2.080198	194.456346	1979.553399	2561.208726	4592.482917
2.086448	197.395085	1973.707201	2649.111868	4072.683991
2.092698	193.306442	2140.341610	2862.463672	4177.193794
2.098948	189.794560	1849.949985	2894.378808	4297.682683
2.105198	197.129765	1829.240602	3007.350315	4331.259902
2.111448	204.193631	1814.348616	3093.772711	4395.614240
2.117698	204.897365	1741.488626	3056.633906	4341.903574

/e/ (the vowel in "Repaka")

Time_s	F1_Hz	F2_Hz	F3_Hz	F4_Hz
3.762258	627.832100	1788.322050	2488.828173	2910.480164
3.768508	596.248641	1793.611774	2453.862965	2984.023829
3.774758	616.937082	1859.702537	2475.439962	3108.351416
3.781008	634.485372	1886.148096	2519.016520	3121.525969
3.787258	620.792966	1888.665574	2523.932922	3086.244825

3.793508	603.824286	1898.079152	2536.046230	2956.321915
3.799758	582.692766	1887.709390	2487.834954	2993.194800
3.806008	564.134291	1902.022791	2475.856950	3272.399793
3.812258	567.892618	1920.132740	2432.472141	3056.409030
3.818508	561.706642	1904.820383	2400.224411	3139.085744
3.824758	557.425190	1880.269024	2427.733523	3349.691430
3.831008	550.747744	1828.669878	2503.867533	3315.994036
3.837258	535.755802	1848.891065	2550.150822	3239.650056
3.843508	538.870932	1910.418063	2551.213478	3342.912609
3.849758	533.656665	1909.299203	2484.952299	3124.955985
3.856008	527.783660	1824.590804	2471.750363	3369.262129
3.862258	526.275127	1772.017606	2416.102535	3460.436072
3.868508	514.595607	1686.206830	2331.605479	3369.992492
3.874758	490.015477	1594.294842	2268.538396	3608.910680
3.881008	465.213203	1482.769620	2242.966174	3631.113442
3.887258	372.012698	1220.641703	2271.645826	3569.758401
3.893508	435.581578	1709.349845	2415.123731	3594.214513
3.899758	356.502293	1735.861151	2481.128039	3637.062554
3.906008	359.394112	1547.106977	2457.278082	3632.467137
3.912258	305.499488	1538.485342	2510.518583	3585.678774
3.918508	271.418374	1507.187867	2572.784684	3544.414342
3.924758	260.450166	1499.142946	2414.939664	3172.030717
3.931008	219.970582	1673.262353	2350.186960	3505.359011
3.937258	1391.159238	2541.587277	3554.459025	--undefined--
3.943508	1086.930916	2522.667970	3531.480736	4054.861015

/a/ (the vowel in "Repaka")

Time_s	F1_Hz	F2_Hz	F3_Hz	F4_Hz
4.016939	676.148744	1218.205946	2539.897636	3730.373088
4.023189	708.350541	1260.956899	2479.879673	3848.828749
4.029439	753.313409	1296.580575	2484.539703	3892.249818
4.035689	792.196871	1356.352954	2512.816621	3838.921063

4.041939	800.542085	1321.404511	2490.913620	3712.605951
4.048189	795.571011	1324.965068	2498.085487	3715.411010
4.054439	784.836024	1302.033658	2512.765708	3656.542952
4.060689	807.727377	1285.097857	2519.583783	3729.420577
4.066939	795.157025	1288.625802	2503.765248	3722.612661
4.073189	778.640605	1278.904850	2507.473676	3692.081312
4.079439	767.698673	1270.444132	2498.916006	3651.267970
4.085689	771.646458	1285.273422	2511.435230	3687.796457
4.091939	733.767384	1329.243844	2494.888620	3699.577582
4.098189	700.330443	1348.979187	2494.139488	3683.769598
4.104439	704.446702	1329.994715	2513.370308	3770.266084
4.110689	624.063902	1340.373590	2504.407070	3849.695589
4.116939	595.075694	1328.848919	2516.801391	3733.763331
4.123189	565.555880	1349.532887	2545.137838	3740.728849
4.129439	539.147662	1337.878672	2506.297400	3807.197352
4.135689	436.126131	1298.051775	2518.506671	3832.013590
4.141939	539.058800	1276.273367	2485.566143	3831.632564
4.148189	598.306187	1324.436653	2470.048925	3595.141790
4.154439	577.259686	1337.751187	2501.928650	3489.116413
4.160689	515.272054	1345.999462	2604.758279	3857.733602
4.166939	547.043794	1365.851670	2663.313338	4114.130619
4.173189	1245.915320	2761.178546	4159.437694	--undefined--
4.179439	427.922861	1291.044690	2775.077277	4043.273790

1st /a/ (the vowel in "Prakash")

Time_s	F1_Hz	F2_Hz	F3_Hz	F4_Hz
5.360213	660.504562	1620.152935	2221.663298	3607.006565
5.366463	660.821206	1694.833160	2223.179371	3615.318544
5.372713	659.161314	1718.645168	2206.667752	3631.728800
5.378963	650.286165	1726.756743	2179.929125	3638.174616

2nd/a/ (the vowel in "Prakash")

Time_s	F1_Hz	F2_Hz	F3_Hz	F4_Hz
5.597713	745.252989	1252.132668	2399.999149	3450.453122
5.603963	766.203225	1245.875056	2382.706225	3459.052607
5.610213	762.346631	1248.906126	2358.171543	3448.492315
5.616463	753.477403	1258.539549	2349.842295	3458.822845
5.622713	760.708413	1267.560777	2308.280996	3467.043424
5.628963	749.478534	1287.619333	2246.180931	3471.296603
5.635213	730.582536	1310.605871	2241.873808	3482.510768
5.641463	743.546732	1335.633370	2259.299256	3508.002505
5.647713	754.004950	1371.015858	2289.536729	3517.804134
5.653963	785.199249	1440.623883	2324.313135	3518.896087
5.660213	808.379037	1484.272454	2342.136908	3509.590477
5.666463	736.173473	1519.429746	2393.205293	3582.401171
5.672713	664.440745	1521.827207	2378.817320	3607.474694
5.678963	625.871700	1579.538426	2388.003916	3612.609538
5.685213	1471.240180	2463.271784	3657.077590	5193.869829
5.691463	1390.945718	2642.190060	3750.728160	4901.896237
5.697713	1326.073258	2565.866375	3670.081147	4736.636825